Question Answering for Decision-making in Green Building Design

A Multimodal Data Reasoning Method Driven by Large Language Models

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ABSTRACT

In recent years, the critical role of green buildings in addressing energy consumption and environmental issues has become widely acknowledged. Research indicates that over 40% of potential energy savings can be achieved during the early design stage. Therefore, decision-making in green building design (DGBD), which is based on modeling and performance simulation, is crucial for reducing building energy costs.

However, the field of green building encompasses a broad range of specialized knowledge, which involves significant learning costs and results in low decision-making efficiency. Many studies have already applied artificial intelligence (AI) methods to this field.

Based on previous research, this study innovatively integrates large language models with DGBD, creating GreenQA, a question answering framework for multimodal data reasoning. Utilizing Retrieval Augmented Generation, Chain of Thought, and Function Call methods, GreenQA enables multimodal question answering, including weather data analysis and visualization, retrieval of green building cases, and knowledge query. Additionally, this study conducted a user survey using the GreenQA web platform. The results showed that 96% of users believed the platform helped improve design efficiency. This study not only effectively supports DGBD but also provides inspiration for AI-assisted design.

1 Conceptual illustration of the Question Answering Platform GreenQA for Decision-making in Green Building Design.

INTRODUCTION

The building industry accounts for up to 40% of global greenhouse gas emissions, making a significant contribution to energy consumption (Wong and Zhou 2015). During the entire lifecycle of a building, the early design stage has the potential to achieve over 40% energy savings (Han et al. 2018). Therefore, promoting decision-making in green building design (DGBD), which focuses on performance-oriented building design, is an important method to deal with climate change and energy shortages (Doan et al. 2017; Y. Zhang et al. 2017).

In traditional DGBD, designers primarily rely on design software for modeling and simulation software for assessment. This process depends on various complex performance parameters and extensive experience, requiring high learning costs. Additionally, the separation between simulation and optimization processes forces designers to repeat the simulation process and manually find the optimal solutions. Consequently, traditional DGBD is inefficient, prone to decision fatigue (Attia, El-Degwy, and Attia 2024) and lacks generalization capability in multiple scenarios (Wang, Rivard, and Zmeureanu 2005). To address these issues, previous studies have leveraged artificial intelligence (AI) algorithms (Debrah, Chan, and Darko 2022) and rapid performance analysis tools (Lin et al. 2021) to improve the efficiency of the design and simulation process. The development of large language models (LLMs) in recent years offers a new direction. The "question answering (QA)" based on natural language creates possibilities for knowledge and case retrieval (Petroni et al. 2019), providing new ideas for the promotion and generalization of DGBD. The field of architecture is also anticipating domain-specific research based on LLMs. However, some argue that LLMs are not adept at data analysis and reasoning, with their scientific accuracy and reliability being widely questioned (Lewis et al. 2020).

To address the issues of high learning costs and the scientific accuracy of data reasoning, as well as to further enhance design efficiency and generalization capability in the DGBD process, this study proposes an integrated question answering framework for multimodal data reasoning, named GreenQA. Based on a multimodal green building knowledge base, GreenQA leverages LLMs through new technical methods. It enables multi-turn interactive QA for various application scenarios, such as weather data analysis and visualization, case retrieval, and knowledge query. This study explores the potential applications of LLMs in the field of green building design, providing new research ideas and a technological approach to the integration of DGBD and LLMs in the future.

STATE OF THE ART

AI in DGBD

In recent years, artificial intelligence and machine learning have been extensively applied in the field of green building (Debrah, Chan, and Darko 2022), supporting green building design decision-making (Petrova et al. 2019; Li, Du, and Kumaraswamy 2024).

Case-Based Reasoning (CBR) methods are effective for retrieving knowledge from existing case libraries to improve decision-making efficiency (T. Liu et al. 2024), but current databases have limitations in both quantity and quality (Li, Du, and Kumaraswamy 2024), which makes it difficult to generate precise generalizations. Various methods have been explored with CBR, such as nonlinear Local-Global retrieval using Artificial Neural Networks to CBR retrieval (Cheng and Ma 2015), text mining integration (Shen et al. 2017), and the application of CBR in building renovation (X. Zhao et al. 2019), and decision-making of building envelopes design during the preliminary design stage (B. Zhang et al. 2021). Additionally, T. Liu et al. (2024) combined the Random Forest (RF) algorithm with CBR methods to address case modification issues.

Natural language processing (NLP) has also been widely applied in green building, though most studies focus on users' perceptions and evaluations of green buildings, lacking the development of decision-making systems. X. Liu and Hu (2019) analyzed Sina Weibo data to study public attention to green buildings, while Guo et al. (2021) examined the effect of green building rating certifications on improving public satisfaction with apartments. Additionally, Du et al. (2021) combined NLP with green building rating systems for automatic project classification and evaluation. In the domain of building regulations, Zhong et al. (2020) integrated NLP and information retrieval to construct a building regulation QA system, but a similar integrated platform has yet to be developed in the green building field.

QA by LLMs

Among artificial intelligence methods, LLMs have demonstrated significant efficacy in handling complex natural language understanding and generation tasks, greatly expanding the application scope of natural language processing (NLP) (Touvron et al. 2023; W. X. Zhao et al. 2023; OpenAI et al. 2024). These models, pretrained on large-scale datasets, can generate coherent and relevant text responses, and can perform NLP tasks through zero-shot or few-shot learning when given appropriate prompts (Brown et al. 2020; Chowdhery et al. 2023).

Question Answering (QA) is a core application area of NLP,

aimed at providing accurate answers to user-posed questions. Traditional QA systems typically rely on structured knowledge bases (Berant et al. n.d.) and complex query mappings (D. Chen et al. 2017) to respond to user inquiries, requiring precise NLP techniques (Z. Li et al. 2024). With the development of LLMs, researchers have begun to explore the potential of using LLMs as knowledge bases for QA (Petroni et al. 2019), and to compare them with traditional knowledge-based QA models (Payne et al. 2023).

Although LLM-based QA systems can perform reasoning on open-ended questions, general LLMs often lack specialized knowledge in the field of green building. This can lead to the generation of answers that are coherent in natural language but factually incorrect (Kim et al. 2023; Zhuang et al. 2023), known as hallucination (Ji et al. 2023; Welleck et al. 2019). This limitation significantly constrains the support that LLMs can provide in the field of green building.

METHODS

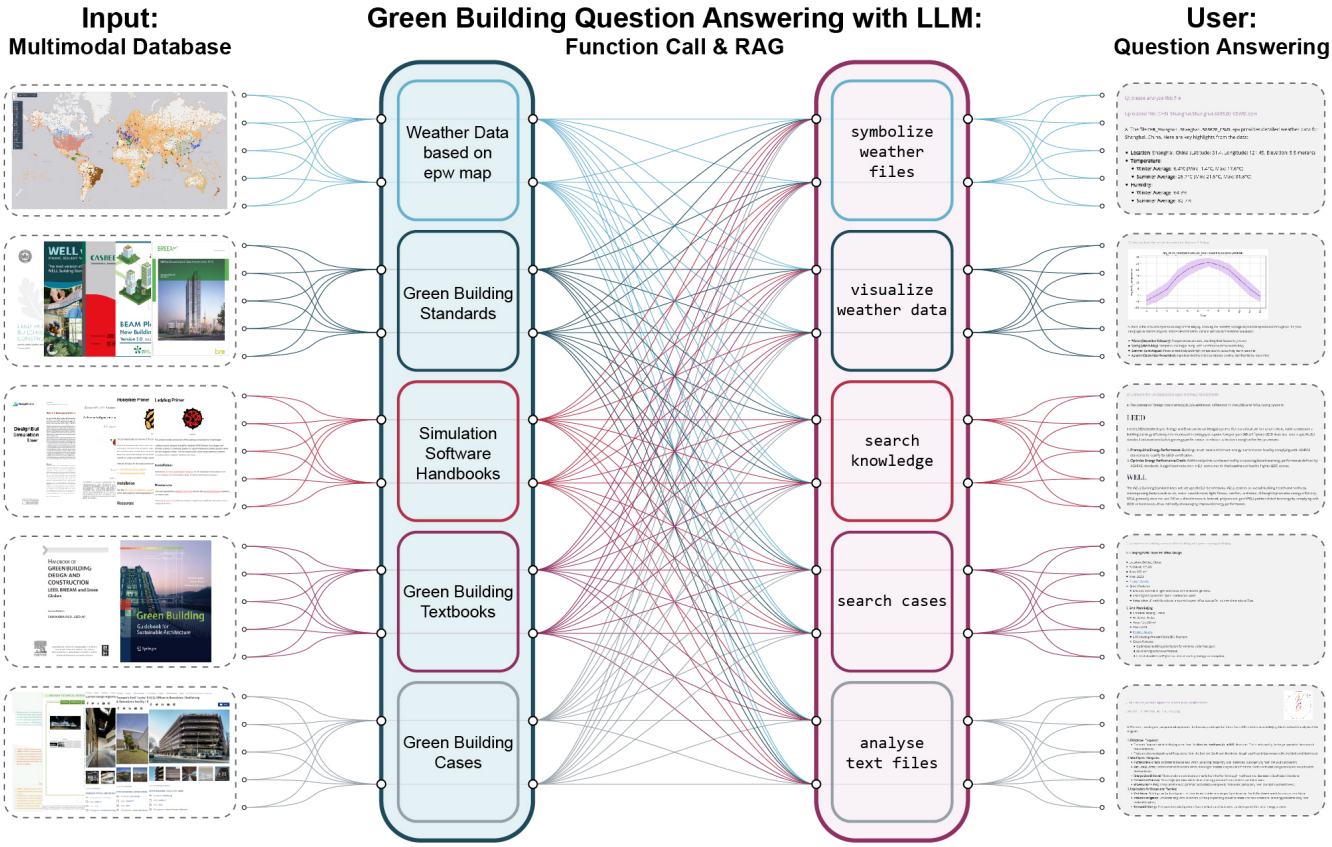
This study developed the GreenQA platform, which supports knowledge retrieval and design interaction in green building design. The technical framework of the platform is illustrated in Figure 2, and its construction is based on a multimodal green building knowledge base.

To address various QA needs in DGBD scenarios, this study enhances the performance of LLMs in three aspects and restructures the output response mechanism. First, by using Retrieval Augmented Generation (RAG) to automatically connect the knowledge base and prevent hallucinations. Second, by employing Chain of Thought (CoT) to enhance LLMs' contextual memory and reasoning capabilities. Third, by utilizing Function Call to integrate LLMs with external interfaces, based on the previous two technologies, to meet the demands of more complex data reasoning tasks and support the reading and generation of documents, images, tables, and other modalities.

Integrating these technical methods, an online QA platform, GreenQA, was developed on the web. A user survey was conducted to evaluate its performance and identify areas for improvement.

Knowledge Base Preparation

DGBD relies on multimodal background knowledge, encompassing technical indicators like acoustics, lighting, and thermal performance, which have led to the creation of numerous textbooks and standards. Globally, up to 600 green building rating systems have been established to assess sustainability (Doan et al. 2017; Y. Zhang et al. 2017;



2 Research Framework.

Multimodal Green Building Knowledge Base

Category	Description	Content	Source
Textbooks	Green building related teaching textbooks	Green Building: Guidebook for Sustainable Architecture  Handbook of Green Building Design and Construction 	Green Building: https://link.springer.com/book/10.1007/978-3-642-00635-7 Handbook of Green Building Design and Construction https://www.sciencedirect.com/book/9780128104330/handbook-of-green-building-design-and-construction
Green building Standards	Technical manuals and rating systems for globally recognized green building evaluation standards	BREEAM  LEED v4.1  WELL v2  CASBEE  GBC-BEAM Plus 	BREEAM: https://www.ibeam.hk/public/knowledgeData-base/?tab=downloadArea LEED v4.1: https://www.usgbc.org/leed/v41 WELL v2: https://v2.wellcertified.com/en/wellv2/resources CASBEE: https://www.ibecs.or.jp/CASBEE/english/downloadE.htm GBC-BEAM Plus: https://www.ibeam.hk/public/knowledgeData-base/?tab=downloadArea
Simulation Software Handbooks	User manuals and documentation for widely used building performance simulation softwares	Design Builder DeST EnergyPlus Ladybug Tools OpenStudio 	Design Builder: https://designbuilder.co.uk/download/documents DeST: https://www.dest.net.cn/zlzx EnergyPlus: https://energyplus.net/documentation Ladybug Tools: https://www.ladybug.tools/resource.html OpenStudio: https://nrel.github.io/OpenStudio-user-documentation/
Green Building Cases	A total of 1,200 project cases certified by globally recognized green building evaluation standards including LEED, BREEAM, CASBEE and GBC-BEAM Plus	LEED BREEAM CASBEE GBC-BEAM Plus 	LEED: https://www.archdaily.com/ BREEAM: https://bregroup.com/products/breeam/breeam-case-studies/#case CASBEE: https://www.ibecs.or.jp/jsbd/ GBC-BEAM Plus: https://greenbuilding.hkgbc.org.hk/projects/list_view

3 Multimodal Green Building Knowledge Base.

Awadh 2017). Notably, the UK's BREEAM and the US's LEED standards have been pivotal in advancing sustainable buildings (Lee 2013). Research indicates that BREEAM and LEED-certified buildings consume approximately 30% less energy than non-certified buildings (Doan et al. 2017). The application of these evaluation standards has accumulated numerous cases and practical experiences, which can be leveraged to address DGBD challenges.

Based on this logic, we constructed a multimodal green building knowledge base comprising four components: green building theoretical textbooks, green building rating standard guidelines, performance simulation software manuals, and green building cases. The specific contents of the knowledge base are illustrated in Figure 3. The books, standards, and manuals are sourced from their respective official websites.

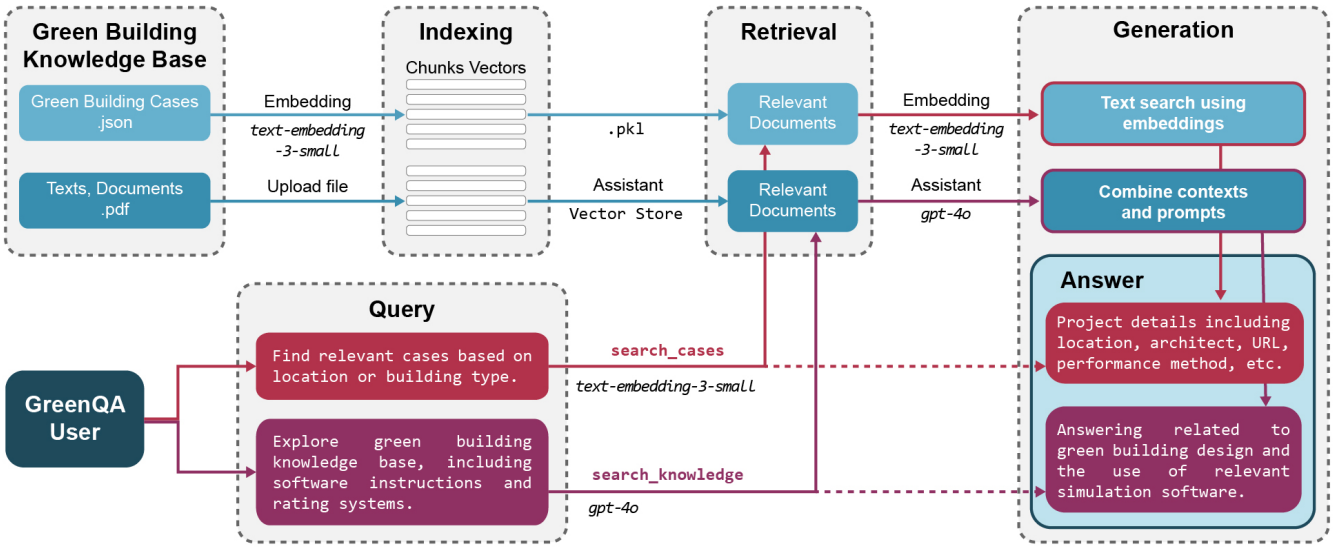
The green building cases include certified projects from four green building rating systems, taking the collection and cleaning of LEED cases as an example. First, the process begins with using web scraping tools to gather basic information and text descriptions of LEED-certified projects from the ArchDaily website. Next, the OpenAI API is called to process the natural language in the case texts, extracting sentences related to LEED ratings and building performance. The project information and relevant descriptions are then organized into structured JSON files.

QA by RAG

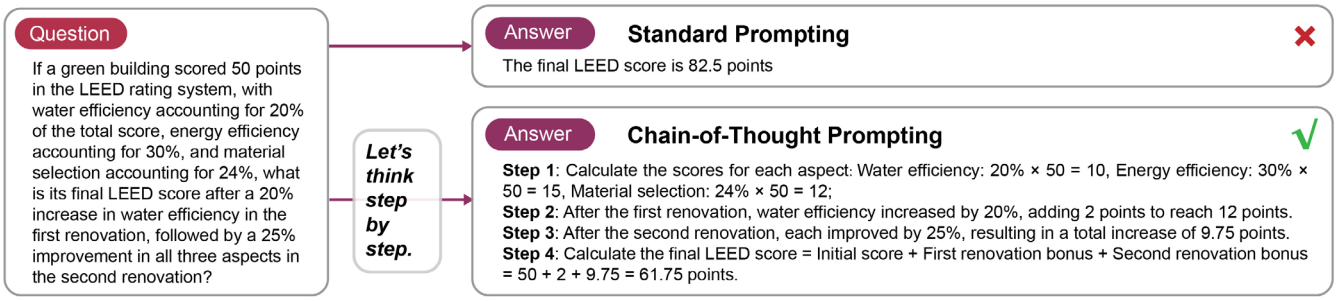
Due to the lack of domain-specific knowledge bases in general LLMs, they tend to generate hallucinations or incorrect answers when handling knowledge-intensive tasks. Relevant research has proposed the Retrieval Augmented Generation (RAG) method to improve the accuracy of LLMs in QA tasks (Lewis et al. 2020). In knowledge-intensive industries such as law and medicine, studies have already demonstrated the use of the RAG method to provide more accurate answers with the aid of knowledge bases (Louis, Dijck, and Spanakis 2024; Singhal et al. 2023). However, in the field of architecture, a systematic database and methodology have not yet been developed.

To further achieve the customized application of LLMs in assisting DGBD, this study adopts the RAG method. By calculating semantic similarity, relevant information is retrieved from the external knowledge base to enhance the expertise of LLMs in the field of green building, while also ensuring the timeliness of the knowledge base (Guu et al. 2020; J. Chen et al. 2024; Lewis et al. 2020).

The RAG method in GreenQA is based on the basic framework of indexing, retrieval, and generation (Gao et al. 2024), as illustrated in Figure 4. For each user query, the system



4 Implementation of Cases and Knowledge Base Retrieval.



5 Comparison of Results with/without Using Zero-shot CoT.

reads text data stored in the knowledge base as vectors, calculates the vector similarity between the user's question and the indexed resources, and then generates a response based on LLM. Specifically, green building cases are stored locally as vector files through an embedding model, and calculations and retrievals are then performed for each user query. This method significantly reduces the number of tokens required for each QA session while markedly improving response speed.

QA by CoT

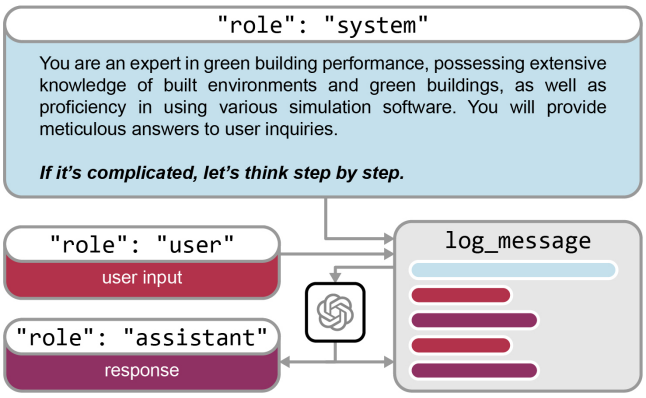
QA tasks for DGBD often involve data reasoning processes with complex interaction logic. Single-turn QA typically results in low answer accuracy for these challenging tasks (e.g., arithmetic and commonsense reasoning). Increasing the model's parameter scale does not significantly enhance performance for such tasks (Lewis et al. 2020). Therefore, we must select a more appropriate technical approach.

In 2022, Wei et al. proposed using Chain of Thought (CoT) to solve this problem (Wei et al. 2022). CoT guides LLMs to break down complex tasks by constructing example prompts, significantly improving task accuracy. This study employs the zero-shot CoT method, incorporating specific instructions like "Let's think step by step" into the prompts (Figure 5). This method is straightforward and achieves significant performance improvements for LLMs with more than 100 billion parameters (Kojima et al. 2022).

To achieve a more continuous interactive experience, we use a loop structure to construct multi-turn conversations, allowing the model to answer subsequent questions based on previous context, which effectively helps users complete complex tasks. This study uses the latest GPT-4o API provided by OpenAI as the QA model (OpenAI 2024). Figure 6 demonstrates the multi-turn QA framework constructed with this model using CoT.

QA by Function Call

In real-world DGBD scenarios, EPW (EnergyPlus Weather) files are crucial for understanding the relationship between buildings and the environment through performance simulation analysis (Crawley et al. 2008). However, directly inputting EPW files and user requests (such as "Please help me analyze this EPW weather data file." or "What type of buildings are suitable for the temperature and humidity in this area?") into LLMs would consume a vast number of tokens, and the probabilistic generation of direct inference results

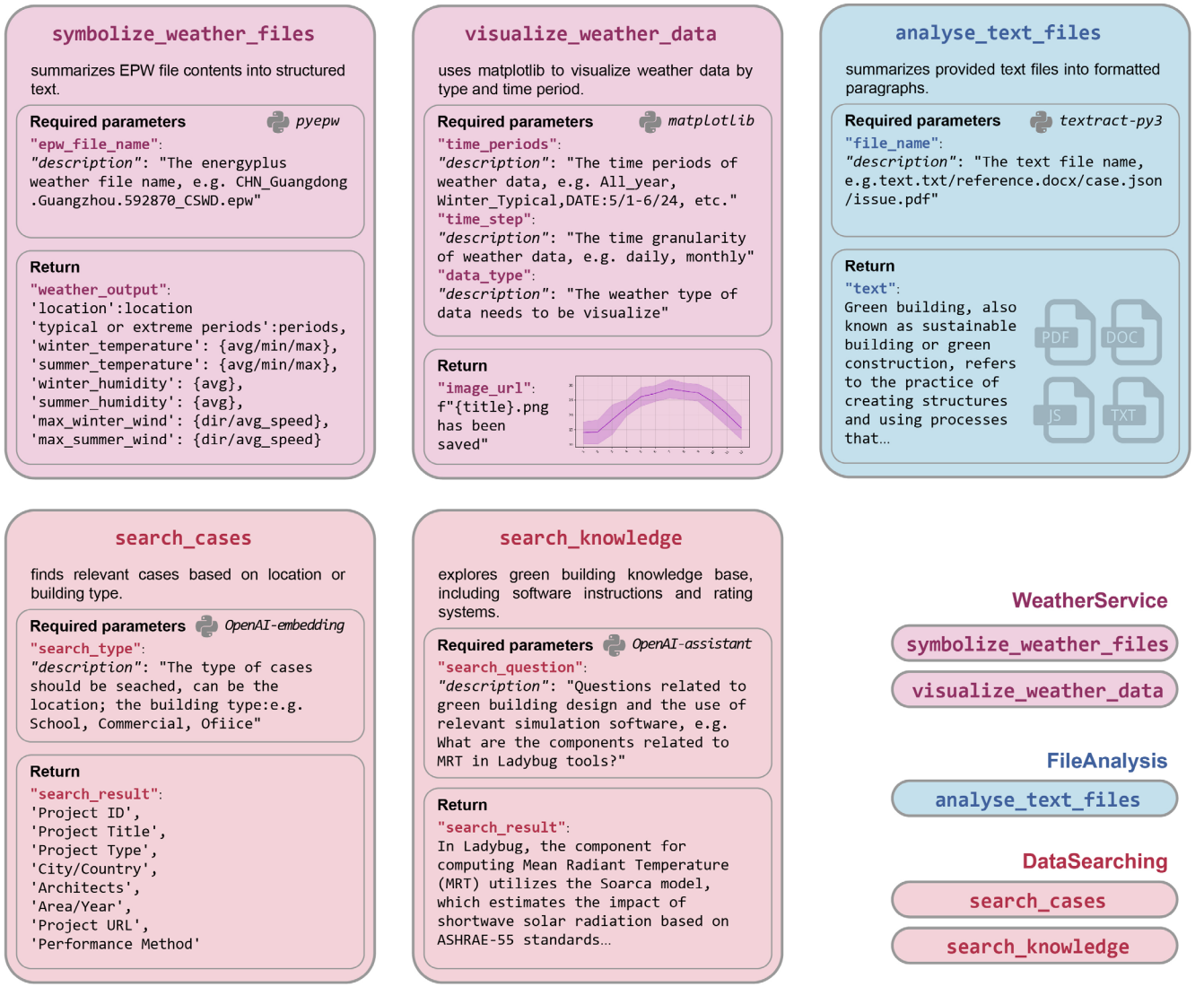


6 Implementation of Multi-turn Dialogue Based on CoT.

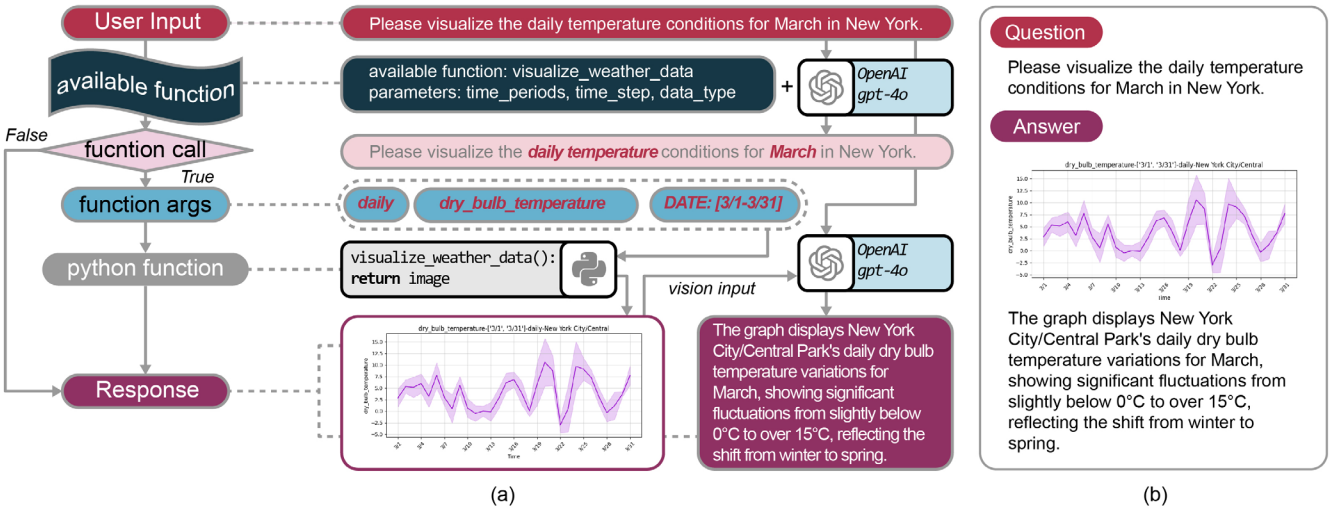
significantly reduces the credibility of data reasoning.

This study uses the Function Call feature to address the above problem. Function Call originated from agent tools in LLM research. In the commonly used LLM application framework LangChain (Harrison 2022), users can create agent scripts based on their needs to extend the LLM's capabilities, particularly its data reasoning functions (Topsakal and Akinci 2023). OpenAI's recent releases have integrated this feature into their API (OpenAI 2023), allowing users to create a function with corresponding input and output descriptions in natural language and a real function script on the back-end. When the LLM processes the user's question, it determines whether to call the function based on the function's description. We have constructed five functions to achieve rich multimodal complex interactions. Figure 7 shows the functions' capabilities and input-output content.

Figure 8 exemplifies the Function Call process using the visualize_weather_data function, demonstrating a potential QA scenario in green building design to better explain the Function Call workflow.



7 Elaboration of Functions and Corresponding Input/Output.



8 (a) Function Call Implementation; (b) Function Call QA Example.

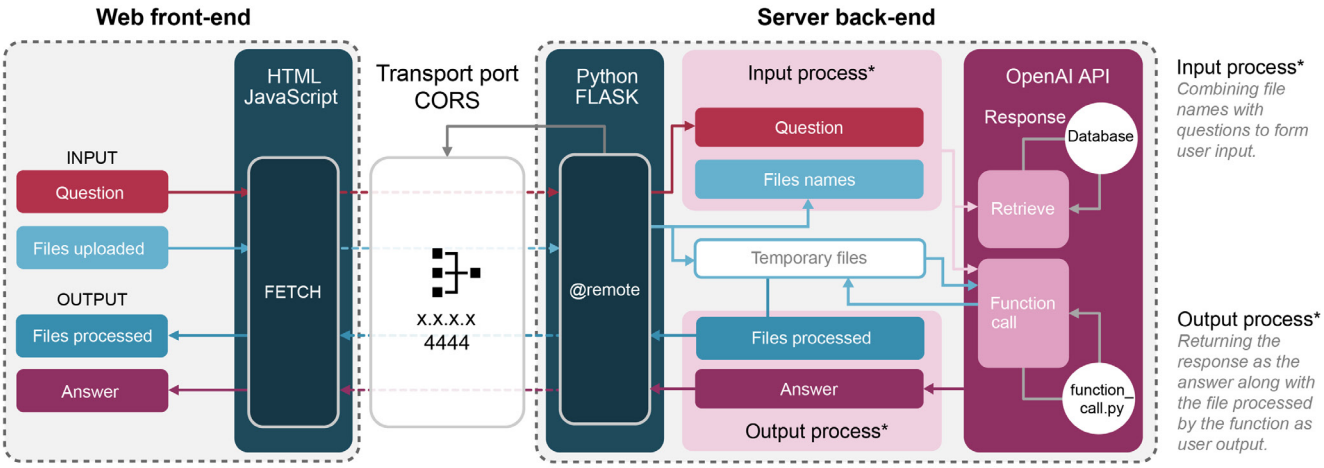
1. Describe the function and its required input parameters in natural language in the tools, extracting the EPW file information as formatted data and using the Matplotlib library to plot meteorological data charts based on user input needs, thereby avoiding direct LLM handling of complex data.
2. Embed a Python function in the back-end that matches the description in the tools for actual data processing, and set it as an executable function in the API's available_functions.
3. Read user input to determine what to call the predefined function, e.g., "Please visualize the daily temperature conditions for March in New York."
4. Interpret the user intent, automatically extract and convert the text to the input format specified for the designed function, such as time_step="daily", time_periods="DATE:3/1-3/31", data_type="dry_bulb_temperature", and pass these parameters to the corresponding Python script.
5. Return the Function Call output, including text and images, as the response, and combine it with the initial query to form a new turn of user input to generate the final response.

Platform Construction

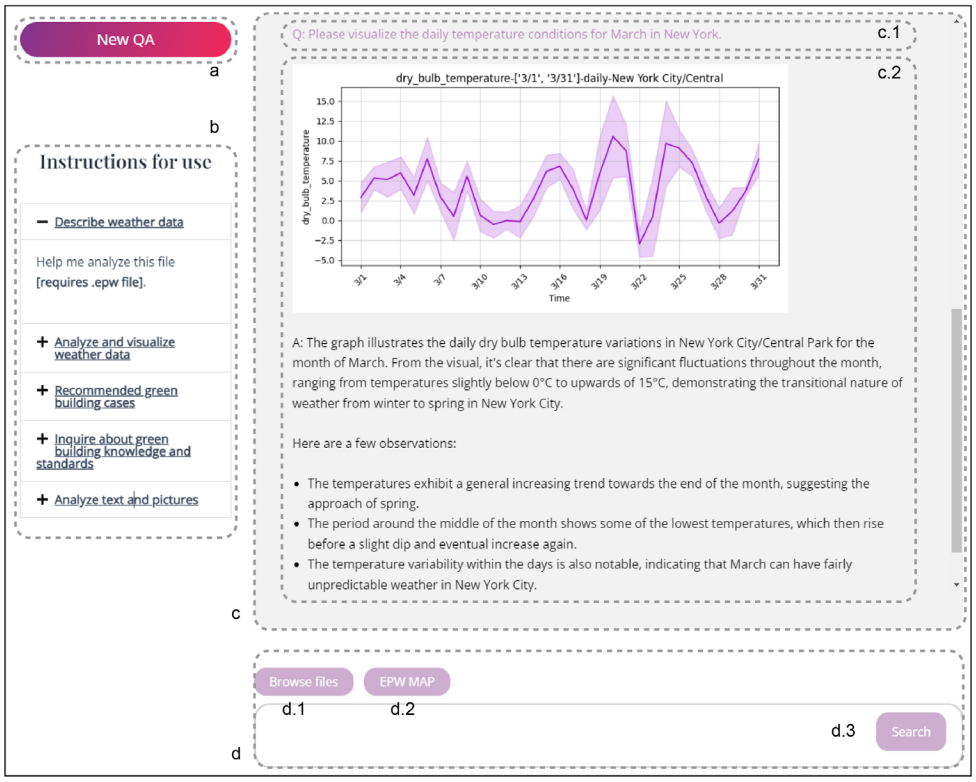
Based on the web platform, this study has developed an interactive website called GreenQA, which uses the technical framework shown in Figure 9 to handle user input and output information flow, enabling knowledge inference QA related to the field of green building.

The back-end of the website utilizes the Fetch mode in JavaScript to send responses, which are received by a Python Flask framework and executed by corresponding Python scripts. Additionally, enabling CORS mode allows these scripts to process cross-origin requests.

The dialog API processing section constructs a complex yet efficient information transmission logic. When the input information flow reaches the back-end, user-uploaded files are saved in a local temporary folder, and the file names are appended to the user input text before being sent to the API as input. This approach allows the Function Call feature to



9 Interaction Logic of GreenQA Platform Front-end and Back-end.



10 Website page layout.

recognize the corresponding file names and call the required functions. When the API generates responses and function calls create charts or other files, these files are separated from the text, saved in the temporary folder, and then sent to the front-end. This method facilitates multimodal interaction and simplifies the recording of information transmission results.

Referring to common QA websites such as ChatGPT, the platform is deployed on a front-end managed by WordPress using HTML and JavaScript markup languages as shown in Figure 10. Users can enter their questions in the search box d.3 located at the bottom of the page. Above the search box, there are two buttons: Button d.1 allows users to upload multiple files for the LLM to read, supporting multimodal file formats such as EPW files, images (.jpeg/.jpg/.png), and text files (.txt/.json/.pdf/.docx). The second button d.2 is used to open the EPW Map global meteorological data visualization website built by Ladybug Tools (Roudsari, Nunzio, and Peng 2023), which helps users find and download EPW data for specific regions. After clicking the search button located at the bottom right, a dialog response will be generated, and the response results will be displayed in the board c above the search box. Clicking button a at the top left of the page, labeled "New QA," will start a new round of dialogue. The lower left corner b provides some QA examples.

User Survey

To better understand user satisfaction and gather suggestions for platform improvement, a survey was conducted, as detailed in the appendix. The questionnaire was designed to evaluate four aspects: multi-turn QA and information interaction, multimodal data and information inference, the value and practicality of key functions, and changes in work efficiency.

The participants in this survey were senior undergraduate and graduate students with previous experience in building performance simulation software or green building design. Before completing the questionnaire, each participant was required to use the platform in a real design scenario for 30 to 60 minutes. Scoring was based on a 5-point scale.

RESULTS AND DISCUSSION

Results of Knowledge Base Construction

The multimodal green building knowledge base has amassed over 3.8 million words of textual materials and includes 1,200 green building cases (Figure 11). These cases span multiple countries and regions across all continents. As shown in Figure 12, the highlighted countries are those with the highest number of cases in each continent.

The type and quantity of projects for green building cases are shown in Figure 13. The main categories include residential,

CATEGORY	NAME	COUNT	
Textbooks	Green Building Handbook of Green Building Design and Construction	88, 015 377, 967	words words
Green building Standards	BREEAM INC Manual	150, 514	words
	LEED v4.1 Guide	335, 005	words
	Well Building Standard v2 Guidebook	134, 537	words
	CASBEE for Buildings (New Construction) Technical Manual (2014)	107, 402	words
	BEAM Plus New Buildings v2.0 (2023)	128, 677	words
Simulation Software Handbooks	DesignBuilder DSM & SBEM Manual v7.2.0	41, 048	words
	DeST Documentation	15, 277	words
	EnergyPlus Documentation	2, 179, 073	words
	Ladybug Tools Documentation	247, 594	words
	OpenStudio Guide	2, 753	words
Green Building Cases	LEED, BREEAM, CASBEE, GBC-BEAM PLus	1, 200	cases
SUM		3,807,862	words
		1, 200	cases

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public, and industrial types. Within the public building category, there are subtypes such as educational, office, and cultural buildings.

Results of Interactive QA

This study uses a simulated interactive case to evaluate the platform's QA results.

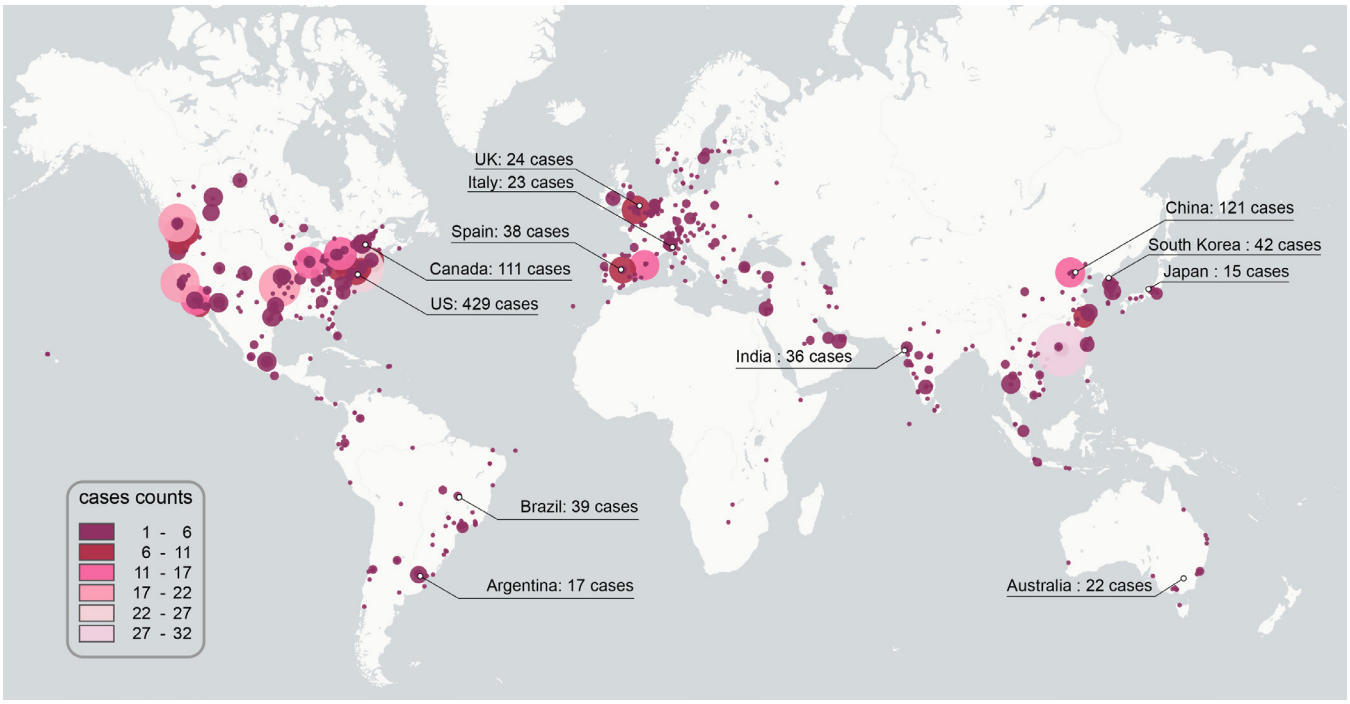
In a simulated design scenario, users engage with the platform to address fundamental questions related to green building concepts, acquire local weather data in EPW format for performance simulations, and integrate data into preliminary architectural designs. The platform assists in extracting key information and generating visualizations, supporting users throughout the modeling and simulation process to address specific design queries and local regulations..

Figure 14 illustrates specific QA content on the platform for various scenarios encountered in the simulated process. It can be observed that, for 1) single-task questions, 2) questions requiring Multi-Function Call, and 3) questions requiring Function Multi-Call with different parameters, this platform consistently delivers professional and rigorous outputs, assisting users in engaging in practical design tasks.

Results of Survey

Figure 15 presents the statistical results of the survey. Among the 25 collected questionnaires, users expressed an overall high level of satisfaction with the GreenQA decision support platform. They particularly appreciated the clear and precise presentation of information as well as the accurate understanding of their needs. Most users found the platform highly valuable for both analyzing meteorological data and retrieving building case studies. Additionally, 96.0% of users reported an increase in work efficiency (scoring above 4) after using the GreenQA platform to assist with DGBD.

However, users gave relatively lower ratings (below 4) for the platform's interaction speed as well as for its robustness and professionalism.

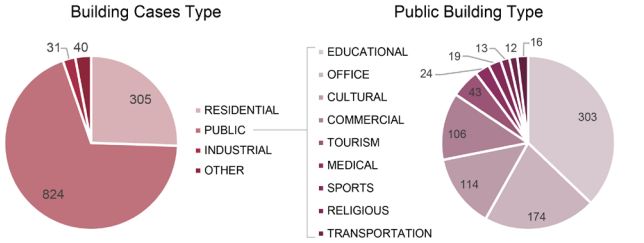


12 Global distribution of green building cases.

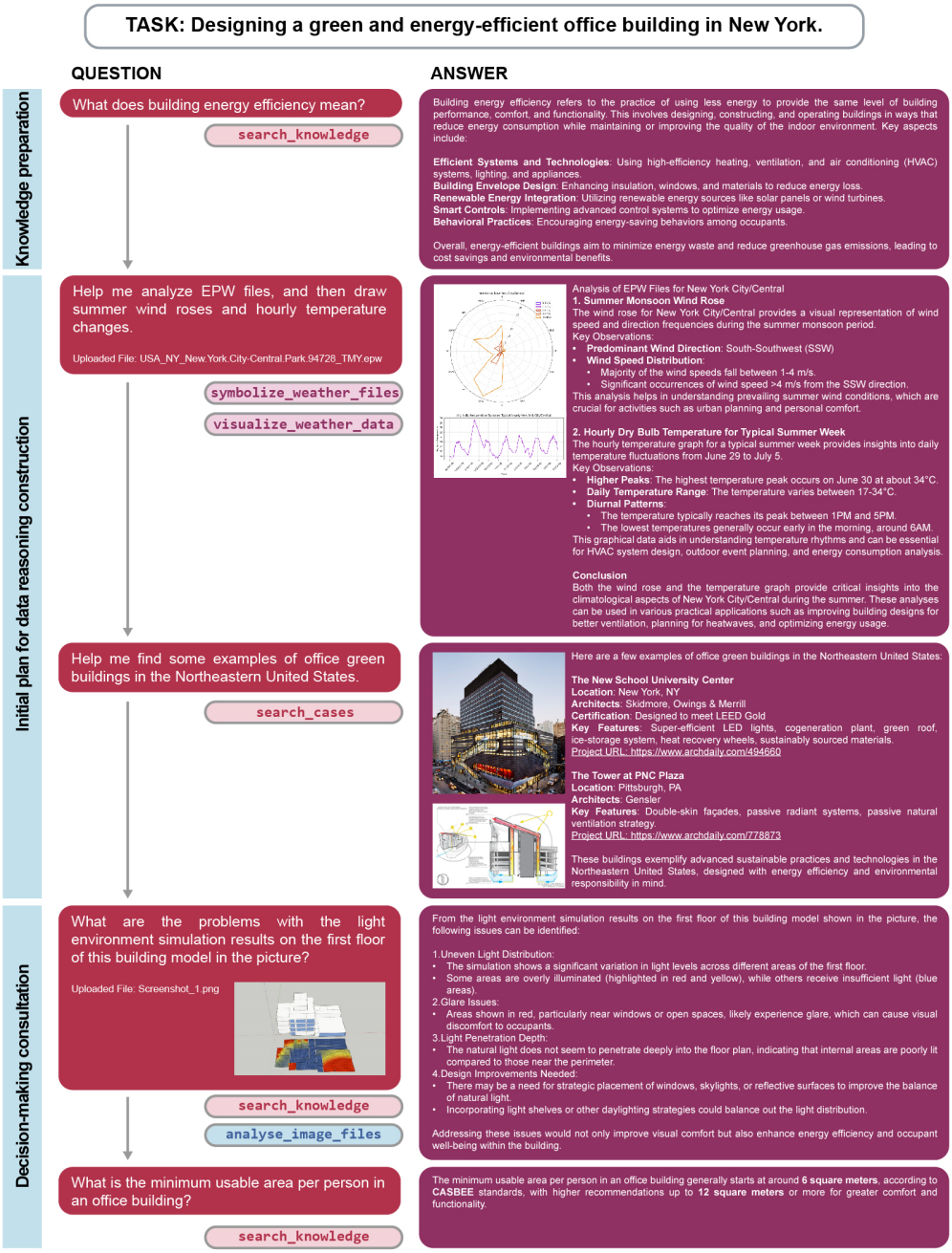
Discussion

This study also compared this platform with other platforms supporting DGBD and weather analysis, such as LEED DSS (Jun and Cheng 2017), ClimaPlus (Arsano and Reinhart 2020), and Gbuilding (Zhen 2024). We examined the input/output content of these platforms and evaluated them based on dimensions such as knowledge base data volume, single input/output response time, and response output mechanisms, as shown in Figure 16.

The comparison shows that the GreenQA platform exhibits significant advantages in terms of functionality, knowledge base size, response time, and interaction methods. GreenQA supports various input and output types, including arbitrary text queries, images, EPW files, and text files, enabling it to



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11 Statistical results of green building knowledge base.

13 Type and quantity of green building cases.

14 A simulation using example of GreenQA platform.

PARTS	EVALUATION CONTENT	RESULTS
Multi-round Q&A and Information Interaction	interface and information presentation	4.08 ± 1.00▲
	interaction speed	3.68 ± 1.11▼
	requirements understanding accuracy	4.16 ± 0.80▲
Multimodal Data and Information Inference	expectations for breadth, specificity, and professionalism	3.80 ± 0.76▼
	requirements for knowledge - compared to ChatGPT	4.36 ± 0.95▲
	requirements for knowledge - compared to search engines	4.16 ± 0.85▲
Practical ranking of main functions	describe weather data	3.12 ②
	analyze and visualize weather data	3.88 ①
	recommend green building cases	3.04 ③
	inquire about green building knowledge and standards	2.88 ④
	analyze text and pictures	2.08 ⑤
Impact on design efficiency	proportion of choosing slightly/markedly increased	96% ▲

15 Statistical results of user survey

Platform	Input	Output	Knowledge base	Response time	Interaction methods
LEED DSS	Formatted-Text inquiry	Formatted-Text results	912 LEED-EB projects	Cases searching: Not provided	ANN-Model for case-based reasoning
ClimaPlus	EPW files	Images of weather data	0	Climate analysis: 5s Indoor simulation :7s	File processing
Gbuilding	EPW files	Formatted-Text results	0	Visualize data: 1.5s LLM evaluation: 3s	File processing LLM
GreenQA	ANY-Text inquiry Images EPW files Text files	ANY-Text results Images of weather data Sheets Hyperlink texts	1200 LEED etc. projects. 3.8M words of green building knowledge	LLM evaluation: 2s Function call LLM: 7-15s	LLM RAG & CoT & Function Call

16 Comparison with other platforms.

handle diverse data types. Additionally, GreenQA possesses an extensive knowledge base containing 1,200 projects and 3.8 million words of green building knowledge, providing comprehensive and accurate information outputs. In terms of response time, it balances flexibility and speed. Furthermore, GreenQA is built on LLM and is compatible with RAG, CoT, and Function Call, offering an innovative and efficient design interaction experience.

Despite positive survey results, some shortcomings remain in the platform's knowledge generalization, interaction performance, and accuracy. Firstly, the green building knowledge base lacks comprehensive coverage, with gaps in green building standards for certain regions, limited image and video materials, and a lack of representative cases from extreme climate zones, such as parts of Africa and the polar regions. Secondly, the platform has relatively low operational redundancy, with API-based language model calls being especially time-consuming when searching the knowledge base. Lastly, although the framework avoids most hallucinations through retrieval augmentation and case-based reasoning, the generative model may still fabricate parameters for Function Call when dealing with more complex and ambiguous questions.

CONCLUSION

As a new paradigm of AI, LLMs are reshaping our work patterns and understanding of the world. Their interaction capabilities are poised to transform design processes, with "Design with LLMs" likely becoming the mainstream approach in computer-aided design in the future. However, significant issues remain, including information accuracy, interpretability, and data security.

This study not only effectively supports DGBD but also redefines traditional design paradigms. Firstly, it establishes a multimodal database containing 3.8 million words of green building knowledge and 1,200 green building cases, filling the gap in large-scale multimodal databases in the field of green building. Secondly, leveraging LLMs' interactive abilities, the study interprets natural language instructions and provides precise responses to DGBD questions. Lastly, by integrating RAG, CoT and Function Call, it enables precise knowledge retrieval and data-model separation, reducing hallucinations and data leakage, and enhancing the professionalism of LLM-based QA in the field of green building.

This study also highlights future directions for the method: 1) localizing large models and expanding the knowledge base, 2) optimizing interaction processes and enhancing search response speed, and 3) increasing fault tolerance for user requirements. This will support more complex and ambiguous design decisions, contributing to energy saving and sustainability in green building design.

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To improve the clarity and grammatical accuracy of the textural content, we utilized the capabilities of ChatGPT [https://chat.openai.com/] for copy-editing purposes only.

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IMAGE CREDITS

All the drawings and images by the authors.

APPENDIX

User Survey Questionnaire

1. *Multi-turn QA and Information Interaction*
    - *Do you find the webpage interface and information presentation of GreenQA clear, concise, and easy to use?*
    - *Are you satisfied with the interaction speed of GreenQA?*
    - *Do you think GreenQA can accurately understand your question requirements?*
2. *Multimodal Data and Information Inference*
    - *Do you think the answers provided by GreenQA meet your expectations for breadth, specificity, and professionalism?*
    - *Compared to directly using ChatGPT, do you think the GreenQA platform can more accurately meet your needs for knowledge about green buildings?*
    - *Compared to manually searching for relevant data, do you think the GreenQA platform can more integratedly and conveniently meet your needs for knowledge about green buildings?*
3. *How would you rank the value and practicality of the main functions of GreenQA?*
    - *Describe weather data*
    - *Analyze and visualize weather data*
    - *Recommend green building cases*
    - *Inquire about green building knowledge and standards*
    - *Analyze text and pictures*
4. *How do you think using the GreenQA platform to assist in green building design decision-making has changed your work efficiency?*
    - *Markedly decreased.*
    - *Slightly decreased.*
    - *Remained the same.*
    - *Slightly increased.*
    - *Markedly increased.*

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